

Fully Automated RNA Extraction and Analyzing Device

Project Overview: This project presents a fully automated RNA extraction and analyzing device that integrates RNA extraction, PCR, and CLIA modules into a single compact system. The device automates the entire process from sample loading to final result generation, minimizing manual intervention and ensuring consistency. It features a single-channel pipette gantry equipped with a gripper capable of lifting and uncapping up to 32 test tubes in one operation. Additionally, the system includes an 8-channel pipette for transferring extracted RNA to the analyzing section depending on the requirement—either CLIA or PCR.

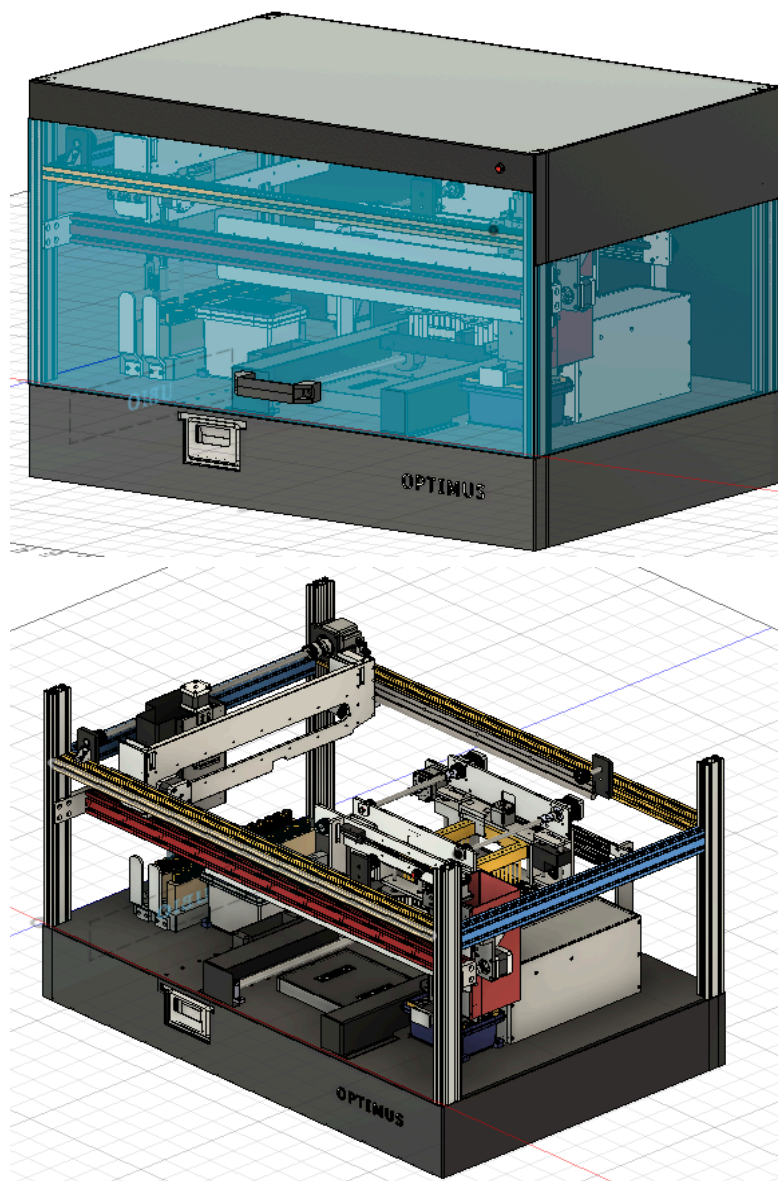
A barcode scanning system is integrated for automatic identification and tracking of each test tube through its serial number. The design also includes an auto-injecting pipette mechanism and an inbuilt waste disposal unit for used pipette tips and residual fluids. The user is only required to place the samples and pipette tips; the remaining operations, including extraction, transfer, analysis, and result generation, are performed automatically. This integration of RNA extraction, PCR, and CLIA within one device significantly reduces time, cost, and operational complexity.

Technical Specifications: -

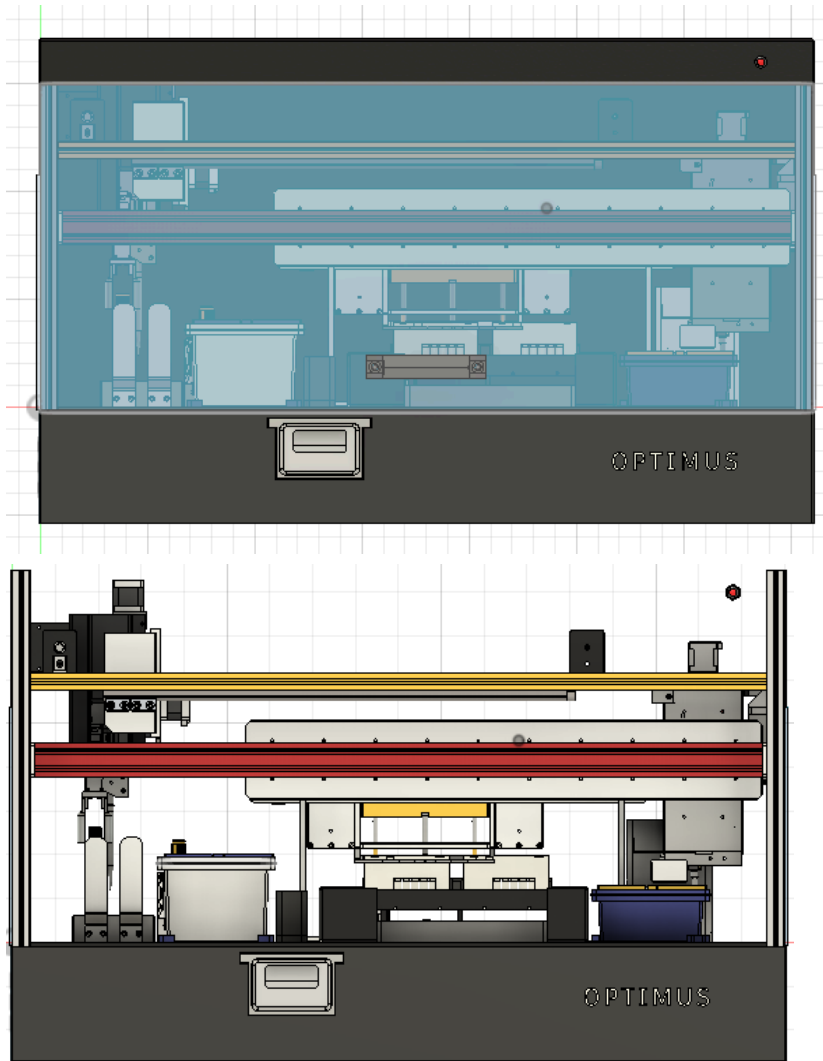
- **Pipetting Gantry:** Single-channel pipette for precision sampling and dispensing.
- **Gripper:** Automated mechanism capable of lifting and uncapping 32 test tubes at once.
- **Secondary Pipette:** 8-channel pipette for RNA transfer to analysis zones (PCR or CLIA).
- **Barcode System:** Integrated high-speed barcode scanner for sample tracking.
- **Motor System:** Moons NEMA 17 stepper motors for precision control and smooth motion.
- **Linear Guides:** Igus sliders for low-friction, accurate linear movement.
- **Waste Management:** Built-in waste disposal for used tips and residual materials.
- **Drive Type:** V-belt driven system for stable, controlled motion.
- **User Interface:** Simple loading mechanism for test tubes and pipette tips.
- **Frame Material:** Aluminum frame for rigidity, durability, and lightweight performance.
- **Pipette Type:** Auto-injecting pipette (single and 8-channel models).

CAD Representation:

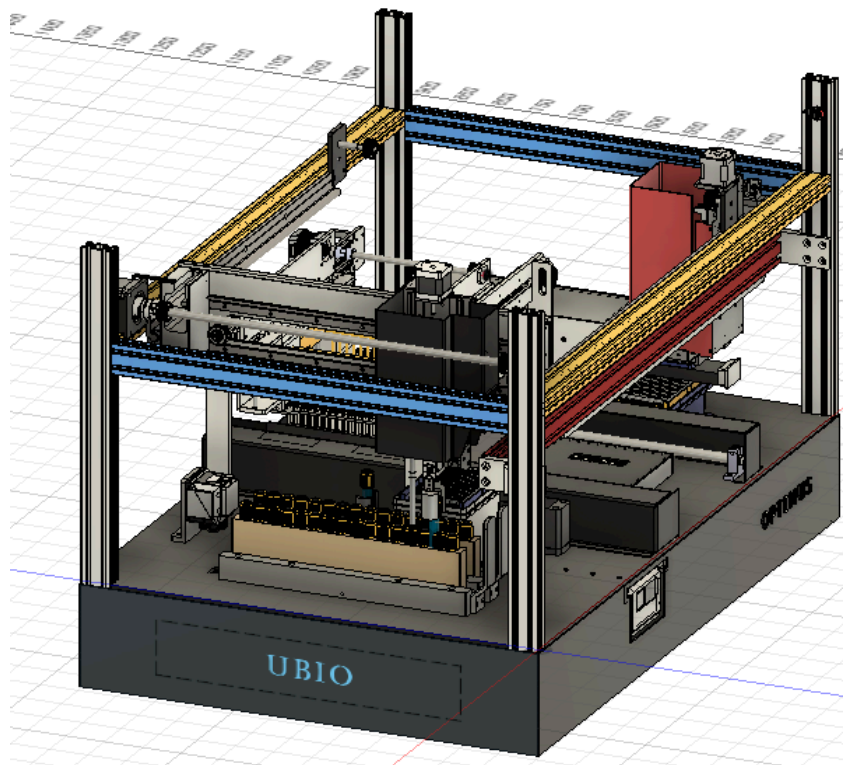
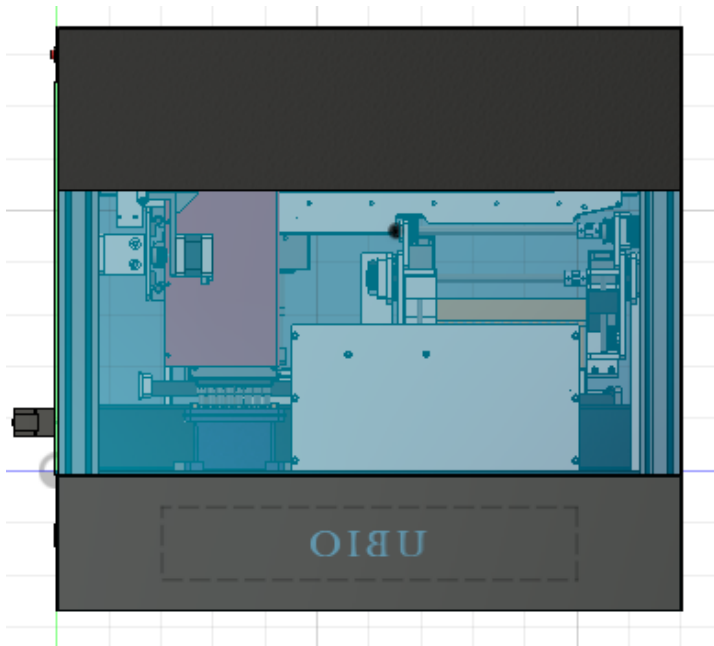
Isometric View: Complete assembly showcasing dual pipette systems and analysis section.



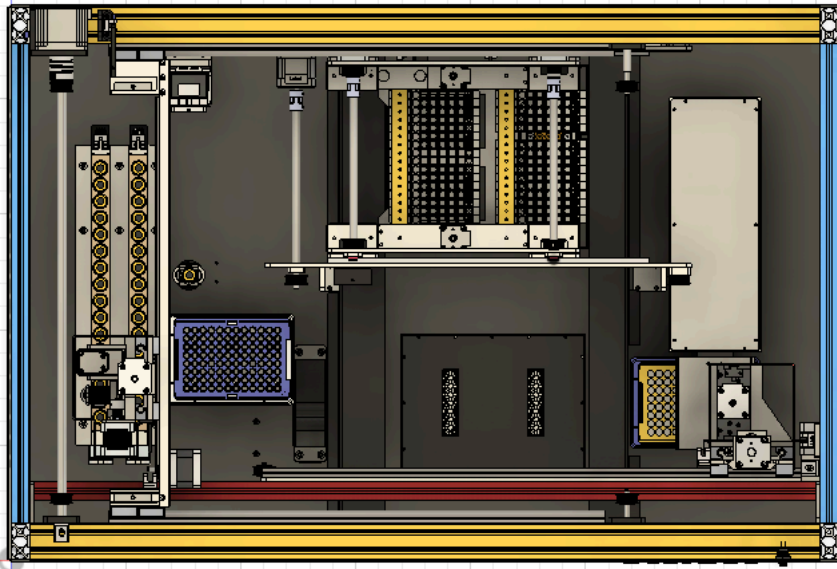
Front View: Demonstrates vertical (Z-axis) pipette and gripper motion.



Side View: Illustrates coordination between pipette, gripper, and analysis unit.



Top View: Displays layout of test tube rack, barcode system, waste unit, and analysis area.



Design Highlights: -

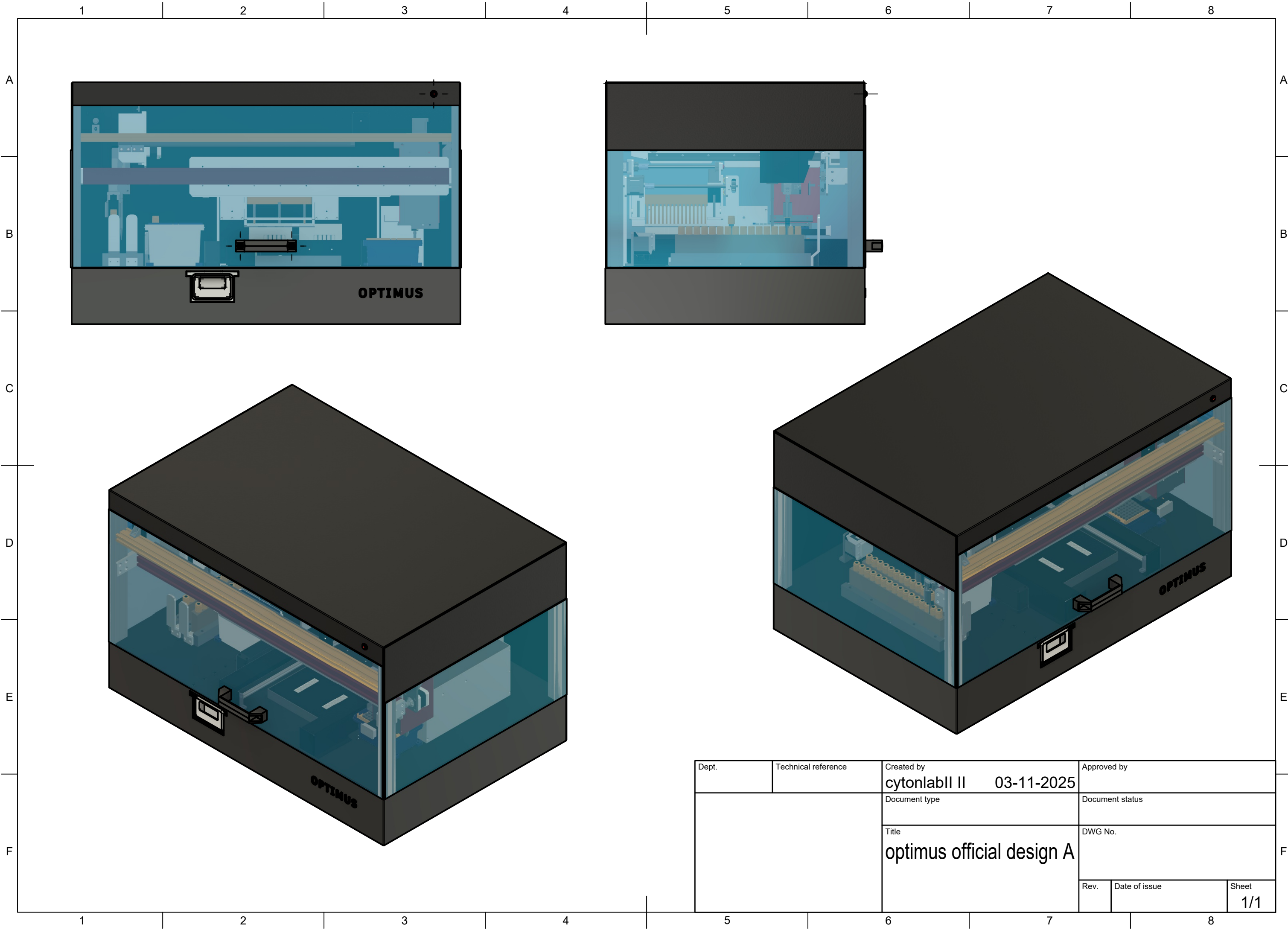
- **Integrated Workflow:** Combines RNA extraction, PCR, and CLIA in a single automated unit.
- **Fully Automated Operation:** Post-loading, all steps are performed automatically.
- **Precision Motion:** Stepper motors and Igus sliders ensure accuracy and repeatability.
- **Smart Sample Tracking:** Barcode scanner maintains sample traceability and prevents mix-ups.
- **Efficient Waste Handling:** Automatic disposal of pipette tips and waste fluids.
- **Dual Pipette Mechanism:** Enables single-tube and high-throughput multi-channel operations.
- **Compact Design:** Space-efficient structure ideal for laboratory automation.
- **Contamination Control:** Uses disposable pipette tips and enclosed pathways to maintain sterility.

Materials & Components: -

- **Frame:** Aluminum – lightweight, rigid, and corrosion-resistant.
- **Linear Guides:** Igus sliders – ensure smooth and precise movement.
- **Motors:** Moons NEMA 17 stepper motors for all controlled motions.
- **Pipettes:** Auto-injecting single and 8-channel pipettes for sampling and analysis transfer.
- **Gripper:** Custom motorized mechanism for uncapping and holding 32 tubes.
- **Couplers / Attachments:** Custom machined components for precision alignment.
- **Drive Mechanism:** V-belt system for reliable, vibration-free motion.
- **Sensors:** Barcode scanner and optical limit sensors for position and identification feedback.

Challenges Faced & Solutions:

- **Challenge:** Synchronizing multi-pipette and gripper operation for multiple tubes.
 - **Solution:** Developed step-sequenced motion control with synchronized motor timing.
- **Challenge:** Compact integration of RNA extraction, PCR, and CLIA.
 - **Solution:** Designed modular structure with shared control and support systems.
- **Challenge:** Preventing sample contamination during extraction and transfer.
 - **Solution:** Implemented disposable tip usage and enclosed pipetting system.
- **Challenge:** Efficient waste disposal management.
 - **Solution:** Added dedicated collection units for both tip and liquid waste.
- **Challenge:** Accurate sample traceability across different modules.
 - **Solution:** Integrated barcode scanning with software-based sample mapping for continuous tracking.



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